Ayush Kumar Singh

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Hostel-10, BIT Mesra, Ranchi (India) - 835215

Education

- **Birla Institute of Technology, Mesra** *Sept 2023 – Sept 2028*
  Integrated M.Sc. in Physics (Minor in Computer Science) Ranchi, India
  - Major CGPA: 9.50 / 10
  - Overall CGPA: 8.83

  Relevant Coursework: General Relativity, Quantum Mechanics (I,II,III), Classical Mechanics, Mathematical Physics, Computational Physics, Statistical Physics, Waves and Optics.

Publications & Conference Presentations

- ***Journal of Astrophysics and Astronomy* (Springer)** *3rd author reviewing draft*
  Manuscript in preparation
  - **Title:** UVIT Study of the Magellanic Clouds (U-SMAC) V. UV Stellar Populations in the Western Outskirts of the Small Magellanic Cloud
  - **Authors:** Ayush Singh, Sipra Hota, Prasanta Kumar Nayak, and Annapurni Subramaniam
- ***Manuscript in Preparation***
  In progress
  - **Title:** Dark Matter–Baryon Interactions in the Hot Halo Gas of the Milky Way
  - **Authors:** Divya Sachdeva, Ayush Singh
- **A Decade of AstroSat Observations: Science Outcomes and Future Prospects** *January 2026*
  Host Organisation: U R Rao Satellite Centre (URSC), Bengaluru, India []
  - Presented a **poster** titled *"Exploring the Western Outskirts of the Small Magellanic Cloud: Tracing Foreground and SMC Stars with UVIT"*, co-authored with Sipra Hota, Prasanta Kumar Nayak, and Annapurni Subramaniam.

Research Experience

- **Institute of Astronomy, University of Cambridge, United Kingdom** *July 2026 – August 2026*
  Title: *Joint Bayesian Inference Pipeline for Dark-Siren Gravitational-Wave Cosmology* **On-site**
  Advisor: Dr. Will Handley and David Yallup
  - Developed a joint Bayesian pipeline combining GW strain and galaxy-catalog likelihoods for precision H_0 inference and scalable dark-siren analyses.
  - Identified and resolved a scalability bottleneck in `CHIMERA`'s batch-based likelihood, enabling its use in gradient-based inference and high-event-rate datasets.
  - Derived a point-wise, differentiable reformulation of the likelihood for direct integration with `BlackJAX`, eliminating reliance on pre-computed GW posterior samples.
  - Gained hands-on experience with HPC/GPU computing by implementing a chunked, GPU-scalable galaxy-catalog likelihood over 22.6M real GLADE+ galaxies, reproducing the unbatched computation to full floating-point precision.
  - Validated the pipeline on real, integrated LVK O1–O4b sensitivity data ($\sim$2.5M injections), enabling application to real GW events.
- **Indian Institute of Technology Hyderabad, India** *May 2026 – July 2026*
  Title:*Dark Matter–Baryon Scattering Constraints from the Milky Way CGM* **On-site**
  Advisor: Dr. Divya Sachdeva
  - Derived upper limits on the dark matter–baryon scattering cross section using the thermal properties of the Milky Way's circumgalactic medium (CGM).

- Investigated four velocity-dependent interaction models ($n = 2, , 0, , -2, , -4$) and compared the resulting CGM constraints with existing CMB limits.
- Demonstrated that CGM-based constraints are comparable to CMB limits for $n = 2$ and $n = 0$, establishing the CGM as a complementary probe of dark matter interactions.
- Developed the analysis and results into a research manuscript, currently in preparation.

- **Indian Institute of Astrophysics, India** *May 2025 - March 2026*
  Title: *Disentangling Stellar Populations in the Western Halo of the SMC Using UVIT and Gaia DR3* **Hybrid**
  Advisors: Sipra Hota and Prof. Annapurni Subramaniam
  - Conducted literature review and processed UVIT and Gaia DR3 data, including photometric calibration and generation of clean catalogs.
  - Applied advanced analysis techniques such as cross-matching, vector point diagrams, GMM, and KDE to study stellar kinematics and separate Milky Way foreground (including 47 Tuc) from SMC members.
  - Constructed UV/optical CMDs and color–color diagrams, performed isochrone fitting, and identified distinct stellar populations. *Manuscript in preparation*

- **Birla Institute of Technology, Mesra, India** *3 months: April 2025 – June 2025*
  Title: *Quasi-Normal Modes of Ellis–Bronnikov Wormholes* **On-site**
  Advisor: Dr. Suman Ghosh
  - Investigated quasi-normal modes of Ellis–Bronnikov wormholes under perturbations.
  - Implemented and compared WKB and Prony methods for QNM computation.

- **Birla Institute of Technology, Mesra, India** *1 year: May 2024 – April 2025*
  Title: *Performance Analysis and Optimization of a Consumption-Scale Biogas Plant* **On-site**
  Advisor: Dr. Rajeev Kumar
  - Led the setup, data collection, and maintenance of a newly built biogas plant.
  - Analyzed system data to optimize efficiency, waste ratios, and mechanical performance.
  - Assisted Bachelor, Master, and Ph.D. students in data analysis and thesis preparation.
  - Developed Project that won an INR 10,000 award at the BIT NISHAN Business Competition.

Workshops and Summer Courses

KSO Summer School, Indian Institute of Astrophysics [🌐] ***10 days:*** *June 2025*
Kodaikanal, India **On-site**

- Participated in hands-on sessions on astronomy, including data cleaning, and basic analysis techniques.
- Attended lectures on different branches of astronomy including general relativity, planetary science, solar studies, stellar evolution, and active galactic nuclei .

Programming & Computational Skills

- **Programming Languages & Tools:** Python, MATLAB, C, C++, LaTeX, Linux Shell Scripting
- **Scientific Computing (Python):** NumPy, SciPy, Pandas, AstroPy, Astroquery, Numerical Techniques, JAX, GPU/CPU-accelerated computing
- **Bayesian Inference:** Hierarchical Bayesian modeling, BlackJAX, MCMC, nested sampling, differentiable probabilistic inference
- **Data Visualization:** Matplotlib
- **Astronomical Data Analysis:** SAO DS9, Aladin Sky Atlas, TOPCAT, IRAF (photometry), DAOPHOT (PSF fitting), CCDLAB (FITS reduction), GLUE
- **Catalogs & Survey Data:** Gaia DR3, Astrosat/UVIT
- **Modeling & Simulations:** Stellar evolution (PARSEC)

Honors and Awards

- **Branch Topper Award, Birla Institute of Technology, Mesra, India**
  Coordinator: Dr. Rajeev Kumar
  - Received a cash prize of INR 15,000 for securing the highest academic rank in the Physics branch during the 2nd–6th semesters, recognizing consistent academic excellence and top departmental performance.

Volunteer Experience

- **Volunteer for Rural Development, Unnat Bharat Abhiyan (UBA)** *August 2024 – December 2025*
  Coordinator: Dr. Rajeev Kumar
  - Contributed to rural development initiatives under Unnat Bharat Abhiyan (UBA), supporting projects focused on infrastructure, sanitation, and education in rural communities.
  - Developed leadership, problem-solving, teamwork, and community engagement skills through active participation in field-based development activities.
- **Freshers' Welcome, Department of Physics** *Coming batch of 2024 and 2025*
  Organizing Committee
  - Led event planning and coordination, managing logistics, student participation, and on-ground arrangements in collaboration with faculty and volunteers.
- **Farewell Event, Department of Physics** *2024 and 2025*
  Organizing Committee
  - Lead planning and coordinating farewell events, managing logistics, activities, and student participation.

Additional Information

Languages:

- Hindi (Native)
- Maithili (Native)
- English (C1)
- German (Beginner)

Interests:

- Wildlife
- Philosophy
- Politics
- Debate